Milestone 1 Summary Report

ClimateScope: Visualizing Global Weather Trends and Extreme Events

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1. Dataset Overview

The dataset used for this project is the **Global Weather Repository** dataset obtained from Kaggle. It contains 126,671 observations and 41 variables representing global weather conditions, air quality indicators, and astronomical information.

Dataset Dimensions:

- **Total Records:** 126,671
- **Total Features:** 41

Data Types:

- 23 Float columns
- 7 Integer columns
- 11 Object (categorical/text) columns

The dataset includes location details, timestamp information, meteorological indicators, air quality metrics, and astronomical attributes.

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2. Data Quality Assessment

A comprehensive data validation process was performed.

Missing Values:

Using df.isnull().sum():

- No missing values were found across any columns.
- All 126,671 records were complete.

Duplicate Records:

Using df.duplicated().sum():

- No duplicate entries were detected.

Data Type Verification:

- Numerical variables were appropriately stored as float64 and int64.
- The last_updated column was converted from object format to datetime format to enable time-series operations.

The dataset exhibited strong structural integrity and required minimal corrective cleaning.

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3. Data Preprocessing & Transformation

Although the dataset required no removal of missing or duplicate entries, preprocessing steps were implemented to prepare it for time-series analysis.

3.1 Timestamp Conversion

The last_updated column was converted into datetime format to allow chronological indexing and temporal aggregation.

3.2 Time-Series Indexing

The dataset index was set to the last_updated timestamp column. This enabled efficient time-based operations such as resampling.

3.3 Monthly Aggregation Using Resampling

Instead of manually extracting year and month fields, monthly aggregation was performed using Pandas' time-series resampling functionality:

- Data was resampled using the 'ME' frequency (monthly).
- Mean values were computed for all numeric variables.
- This produced a monthly averaged dataset suitable for seasonal trend analysis.

This method ensures accurate chronological grouping while maintaining computational efficiency.

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4. Statistical Overview

Descriptive statistical analysis revealed results like:

- Mean temperature (Celsius): 21.525°C
- Temperature range: -29.8°C to 49.2°C
- Broad variability in wind speed and air quality indicators
- Global geographic coverage confirmed by wide latitude (-41.3 to 64.15) and longitude (-175.2 to 179.22) ranges

The dataset provides extensive global environmental coverage.

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5. Output Datasets Generated

The following outputs were produced:

1. Cleaned and validated dataset
2. Monthly aggregated dataset created using time-series resampling

These outputs prepare the data for statistical analysis and visualization in Milestone 2.

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6. Milestone 1 Conclusion

Milestone 1 objectives were successfully achieved:

- ✓ Dataset downloaded and inspected
- ✓ Data integrity validated (no missing or duplicate values)
- ✓ Timestamp converted to datetime format
- ✓ Monthly aggregation performed using time-series resampling
- ✓ Dataset prepared for advanced analytical processing

The dataset is now fully structured and ready for statistical exploration and visualization in the next phase of the project.